

Self-Similarity in Online Networks During Social Movements

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Online platforms provide an infrastructure for social movements, leaving digital traces that can be modelled as networks to quantify how information, participation, and coordination emerge during episodes of collective action and evolve over time. In this work, we unveil the emergence of scale-invariant online interaction patterns in social movements through network analysis of three geographically and sociopolitically distinct massive mobilisation events. By constructing co-occurrence networks from Twitter (now X) hashtag data and applying a degree-thresholding renormalisation procedure, we demonstrate that these highly correlated social phenomena exhibit clear signatures of self-similarity at peak mobilisation times. These critical points are characterised by modular-to-nested transitions, both in the co-occurrence networks and the bi-partite ones, maxima in user participation, and clustering spectrum collapse across multiple network scales. Despite their geographical and sociopolitical diversity, all three movements display remarkably analogous self-similar properties. Furthermore, the results hint at the emergence of a latent metric structure that supports successful hyperbolic embedding, providing an estimate of effective social distance. Together, these findings suggest that self-similarity may constitute a universal organising principle of social movements during peak mobilisation phases, as it lays the groundwork for the rapid amplification of information across scales that is necessary for the successful coordination of collective action.

I. INTRODUCTION

Cooperative social phenomena in online networks are short-lived bursts of coordinated behaviour in which many users align around a shared target or theme. This produces high correlations in activity. Users post and interact in synchronised bursts, reuse the same language, hashtags, memes, and URLs, and engage with the same accounts. This creates overlapping interaction patterns due to the shared goal of attracting attention and achieving a common objective within the short time span of an event. Large-scale studies have connected online social media behaviors to offline protest, showing that increased coordination of messages on social media such as Twitter using specific hashtags is associated with subsequent large-scale decentralised protests [1–3]. In network terms, structural signatures of these episodes include temporally clustered edges, denser and more cohesive communities, greater centralisation around hubs, and more similar user interaction profiles, with the coordination signal fading once the event window closes [4–6].

Intuitively, the structure of online social networks during peak mobilisation phases may suggest the emergence of critical-like states in which collective action becomes highly correlated across users and time [6], so that small triggers can cascade into system-wide bursts of activity. However, these critical-like signatures remain largely uncharacterised across scales. In particular, scale invariance and other statistical scaling properties in multiscale network organisation during these mobilisation episodes remain largely unexplored at the topological level, and the relatively scarce existing work has mostly focused on dynamical characterisation and modelling. [7, 8].

In complex networks, structural scale invariance has been investigated as self-similarity across length scales, adapting Mandelbrot’s ideas to characterise certain geometric objects known as fractals [9, 10], which express the invariance of form under length rescaling. Such rescaling has been implemented in networks using a variety of definitions of internode distance [11]. For instance, shortest path length was originally used in combination with box-counting coarse-graining to explore fractality and define fractal dimensions in networks [12, 13]. However, the shortest path length approach has serious limitations in networks due to the small-world property, which limits

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the range of available shortest path-length scales, hence other approaches have been introduced.

In particular, Serrano et al. [14] found that some scale-free networks are self-similar under a simple degree-thresholding renormalisation scheme that produces a nested hierarchy of subgraphs by progressively filtering out nodes with degrees below a given threshold. This self-similarity implies that the same connectivity law operates at different scales and it is captured by the statistical equivalence of the resulting subgraphs. This symmetry, referred here as multiscale k -cut self-similarity, finds a natural interpretation under the assumption that network nodes exist in a hidden metric space that endows pairs of nodes with meaningful distances [14].

It was later shown that this latent geometry is effectively hyperbolic [15], and a variety of model-based network embedding methods were developed to produce maps of real networks in which the likelihood that two nodes are connected decreases with their hyperbolic distance [16–19], along with data-driven techniques [20]. The geometric approach has enabled the definition of renormalisation transformations [21, 22] which revealed that real networks display self-similar structure across length scales [21], including brain connectomes [23, 24], and during time evolutions [25]. More recently, it has also been shown that high levels of clustering must be accompanied by multiscale k -cut self-similarity for networks to be possibly describable by Riemannian geometric spaces [26], since geometricity within a broad family of random graphs defined in Euclidean, spherical, or hyperbolic spaces implies both clustering and k -cut self-similarity in complex networks.

Motivated by the fact that online social networks are already characterised by high levels of clustering due to the triadic closure mechanism that drives human interactions [27], in this work we explored the complementary condition implied by latent geometry in complex networks. More precisely, we used the degree-thresholding renormalisation procedure to search for signatures of multiscale k -cut self-similarity in the structural features of online social networks in the context of social movements. We found clear evidence of self-similar structural patterns emerging at peak mobilisation times across diverse geographical and sociopolitical contexts, suggesting that multiscale k -cut self-similarity may act as an ordering principle during highly correlated and clustered social events, fostering the emergence of effective social distances.

The remainder of this paper is organised as follows. We continue with section II, where we present the events for which the data were collected, as well as the methodology used to gather those data. Then, in Section III, we report the empirical results and provide a comprehensive analysis of the observed outcomes obtained through the application of the methodology described in the section *Methods*, where we present a detailed exposition outlining the sequential analytical steps and computational procedures employed. Finally, in Section IV, we summarise

the main findings and implications of our work.

II. DATASETS

We analysed data collected from Twitter (now X) covering three geographically and sociopolitically distinct episodes of large-scale mobilisation events. In each case, online activity surged in the days surrounding the event and culminated in street demonstrations. Two datasets correspond to protests in Buenos Aires, Argentina. The first one, *9n*, was held on 9 November 2019 against the government’s proposed judicial reforms (the “*9n* gran marchapor la justicia” protest). The second dataset, *No al tarifazo*, was carried out between 4 and 6 January 2019, in which citizens also expressed discontent with government policies, particularly tax increases and rising costs of essential public services. The third dataset, *Charlie Hebdo*, relates to France following the terrorist attack on the offices of the satirical magazine Charlie Hebdo in Paris on 7 January 2015, which triggered a massive demonstration on 11 January 2015.

For each social movement, we identified the two most frequently used hashtags during the demonstration day(s) and compiled the set of users who posted at least one tweet containing either hashtag during that period. We then collected all tweets posted by these users over a broader observation window. Finally, we extracted from those tweets every hashtag appearing in their messages. Table I summarises the seed hashtags used for data collection, and basic dataset statistics (number of users and number of distinct hashtags).

III. RESULTS

A. Degree-thresholding renormalisation procedure

Once all tweets were collected, we aggregated their associated hashtags into Temporal Windows (TWs), which consist on all the hashtags posted between their start and end times. The TW duration for each dataset was selected to maximise the resolution of the modular-to-nested transition, which defines the critical regime in [6], and has also been characterised through entropic measures in [28]. The modular-to-nested transition is manifested as a sharp decrease in modularity accompanied by an increase in nestedness; see *Methods* for more details and the explanation of how these metrics were measured. We tested aggregation intervals ranging from one to six hours. Based on this exploration, tweets related to the *9n* mobilisation were aggregated into one-hour windows, whereas those associated with the *No al Tarifazo* and *Charlie Hebdo* movements were aggregated into two-hour windows.

For each TW, we constructed weighted bipartite networks linking hashtags and users, where the weight of each edge represents the number of times a given user

TABLE I: Description of the datasets employed.

Dataset	Timespan of Data Collection	Seed hashtags	# hashtags	# users	Country
<i>No al tarifazo</i>	01/01/2019 6h – 07/01/2019, 23h	#noaltarifazo and #ruidazonacional	136433	9670	Argentina
<i>9n</i>	08/11/2019, 6h – 10/11/2019, 20h	#9n and #9ngranmarchaporlajusticia	219488	8022	Argentina
<i>Charlie Hebdo</i>	10/01/2015, 1h – 12/01/2015, 23h	#charliehebdo and #jesuischarlie	185670	16146	France

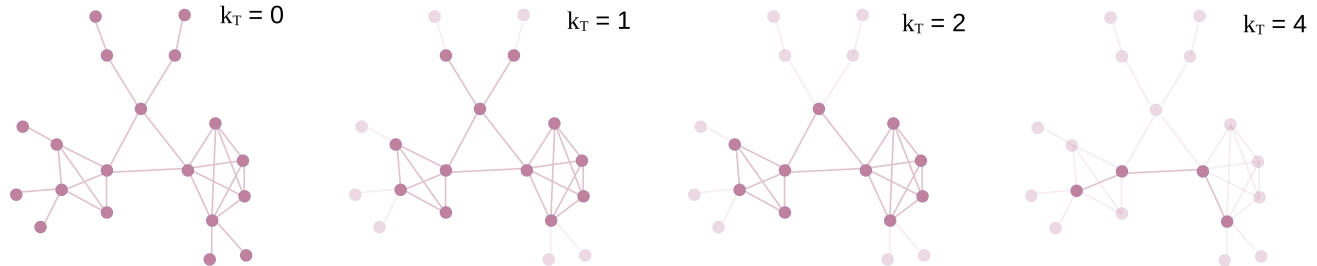


FIG. 1: **Degree-thresholding renormalisation procedure.** We can see the original network ($k_0 = 0$) and three subsequent renormalised ones $k_1 = 1$, $k_2 = 2$ and $k_3 = 4$, where the nodes with k_t links or less were removed from the network at the t -th iteration.

employed the corresponding hashtag during that temporal interval. We also built hashtag co-occurrence networks where hashtags are represented as nodes, and an undirected edge was added between any pair of hashtags co-used by at least one user within the same TW. Edge weights in the co-occurrence network were defined as the number of distinct users who posted both hashtags during that period, thus providing a user-based measure of co-occurrence strength.

The degree-thresholding renormalisation (DTR) procedure, introduced in [14], constructs a nested hierarchy of subgraphs by progressively removing nodes whose degree falls below a given threshold. This hierarchical filtering enables the examination of whether topological properties remain invariant across scales. To quantify the degree of collapse between the original distributions and those of the renormalised subgraphs, we use the ϵ^2 test introduced in [26] (see *Methods* for details). We use both tools below to identify the CTWs and to probe the self-similar structure of the networks.

More specifically, given a graph G , DTR produces a sequence of n subgraphs $G(k_t) \subseteq G$ obtained by removing all nodes from G whose degree is equal to or less than k_t , where $k_t = 0, 1, 2, \dots, n$ denotes the sequence of degree thresholds effectively filtering out less connected elements. Figure 1 illustrates this procedure with a toy example: starting from the original network ($k_t = 0$), nodes with degree k_t or less are progressively removed at each iteration, producing a sequence of subgraphs as k_t increases that retain only the most highly-connected nodes.

B. Detecting the Critical Temporal Window

Figure 2a-c shows the number of unique hashtags and unique users across TWs for the three events. Because these networks reflect online activity within each window, we highlight in gray the intervals between 01:00 and 08:00 local time in the respective countries where each social movement took place. These low-activity TWs contain sparse participation and limited relevant content, and are therefore excluded from further analysis. Within the active periods, there is a single clear dominant peak of engagement in *No al tarifazo*, while *Charlie Hebdo* and *9n* display several maxima associated with moments of maximum online activity.

For each mobilisation, we computed the evolution of modularity and nestedness across all TWs during the period of analysis. The results are reported in Fig. 2d-f for the bipartite user-hashtag networks and in Fig. SI.1 for the unipartite hashtag co-occurrence network. To identify the CTWs unambiguously, we applied the $\epsilon_c^2(k)$ test (see *Methods*), which quantifies the degree of self-similarity across renormalised subgraphs. Fig. 2g-i shows the temporal evolution of $\epsilon_c^2(k)$ for each movement, computed as a moving average over $N = 3$ consecutive TWs to reduce noise. The raw ϵ^2 computed without smoothing are provided in Fig. SI.3 for all three datasets. The unfiltered series exhibit considerable temporal fluctuations, with alternating local minima and maxima that obscure the global structure. This motivates the application of the three-window moving average: by attenuating high-frequency noise, the smoothing procedure enables the robust identification of global minima corresponding to the CTW, without introducing spurious features or obscuring genuine critical windows.

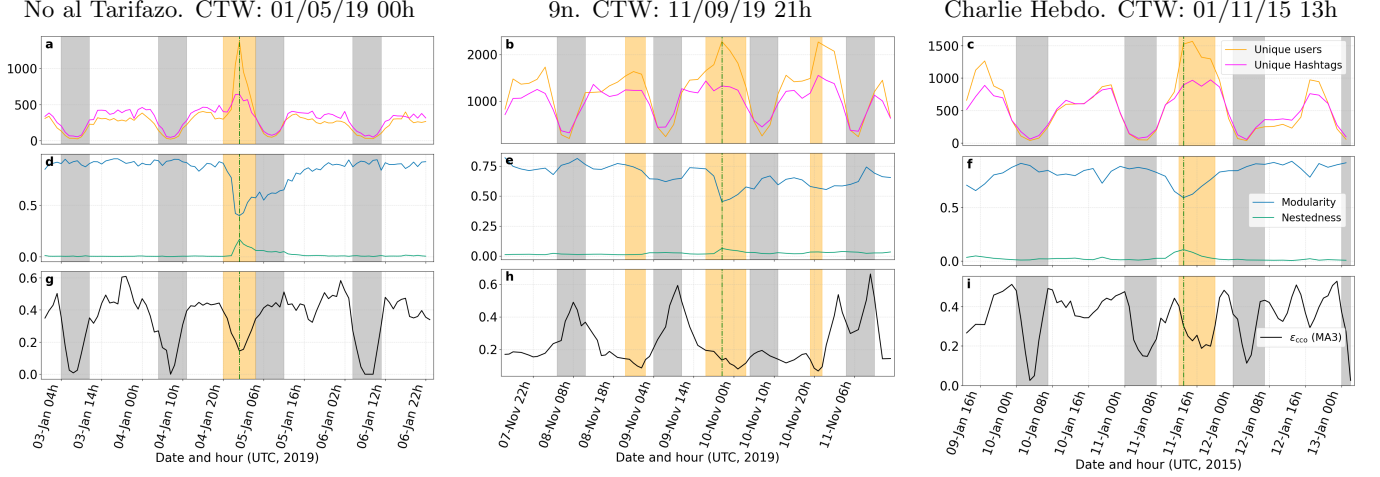


FIG. 2: Detecting the Critical Temporal Window (CTW). (a) Number of unique users and unique hashtags across temporal windows for the three social movements. Grey bands indicate low-activity periods (01:00–08:00 local time) excluded from analysis. (b) Evolution of modularity (blue) and nestedness (pink) in bipartite user–hashtag networks. (c) Temporal evolution of the $\epsilon_c^2(k)$ self-similarity metric, computed as a moving average over $N = 3$ consecutive windows. Yellow bands in all panels highlight intervals with minimum $\epsilon_c^2(k)$ values that coincide with engagement peaks and modular-to-nested transitions, defining the Critical Temporal Windows. The identified CTW for each movement is marked by a black segmented line. For *No al Tarifazo* (left) and *Charlie Hebdo* (right), a single clear CTW emerges. For *9n* (centre), three candidate segments show low $\epsilon_c^2(k)$ values, but only the second exhibits a clear transition in the bipartite networks, unambiguously identifying the CTW. All times are given in UTC.

The minima in $\epsilon_c^2(k)$ identify candidate CTWs. For *No al Tarifazo* and *Charlie Hebdo*, a single clear minimum emerges, which coincides precisely with engagement peaks (Fig. 2g-i) and well-defined modular-to-nested transitions in the bipartite networks (Fig. 2d-f), unambiguously defining unique candidate CTWs highlighted in yellow across all panels.

For *9n*, however, three segments show low $\epsilon_c^2(k)$ values (Fig. 2-h). Notice that unipartite co-occurrence networks (Fig. SI.1) do not help to resolve this ambiguity, as they also exhibit two candidate windows with modular-to-nested transitions, making it difficult to unambiguously identify the CTW. The bipartite networks (Fig. 2h), however, resolve this ambiguity: only the second segment exhibits both a sharp drop in $\epsilon_c^2(k)$ and a clear modular-to-nested transition, unambiguously identifying Saturday, November 9, 2019 from 18:00 to 20:00 (Buenos Aires time, GMT−03:00) as the CTW (marked with a dashed line in all panels of Fig. 2).

For the *No al tarifazo* movement, the transition spans two consecutive TWs in the co-occurrence representation; however, the bipartite networks resolve it into a single, well-defined peak, allowing us to identify the CTW as Friday, January 4, 2019 from 21:00 to 22:00 (Buenos Aires time, GMT−03:00). Finally, using the same reasoning, for the *Charlie Hebdo* dataset, we set the CTW to Sunday, January 11, 2015 from 14:00 to 16:00 (Paris time, GMT+01:00). The bipartite representation thus provides a clearer signal for detecting the CTWs and is used here as the primary guide for discerning the transi-

tions.

C. Multiscale k -cut self-similarity

Next, we provide evidence of multiscale self-similar network connectivity emerging at the identified CTWs across the three datasets. To this end, we applied the DTR procedure to the co-occurrence networks in the selected windows and checked the scaling of their topological network properties, specifically the degree distribution and the clustering coefficient, in the sequence of nested subgraphs generated by the thresholding procedure. Prior to the appliance of DTR, we filtered each network to remove hashtags tweeted by fewer than $N = 3$ users for *No al tarifazo* and *Charlie Hebdo* and $N = 5$ users for *9n*, thereby reducing noise while preserving the core structural features. Table SI.1 presents the percentage of nodes and edges retained after this filtering process for each dataset.

Having identified the CTWs, we now examine whether the network’s structural properties remain invariant across scales using the DTR hierarchical filtering. To quantify the collapse between the original curves and those obtained for the DTR subgraphs, we computed their distance by means of the ϵ^2 test (see Methods for details). At the identified CTWs, the topological properties of the DTR subgraphs display a maximum level of self-similarity, as evidenced by the low ϵ^2 values shown in Fig. 2g-i.

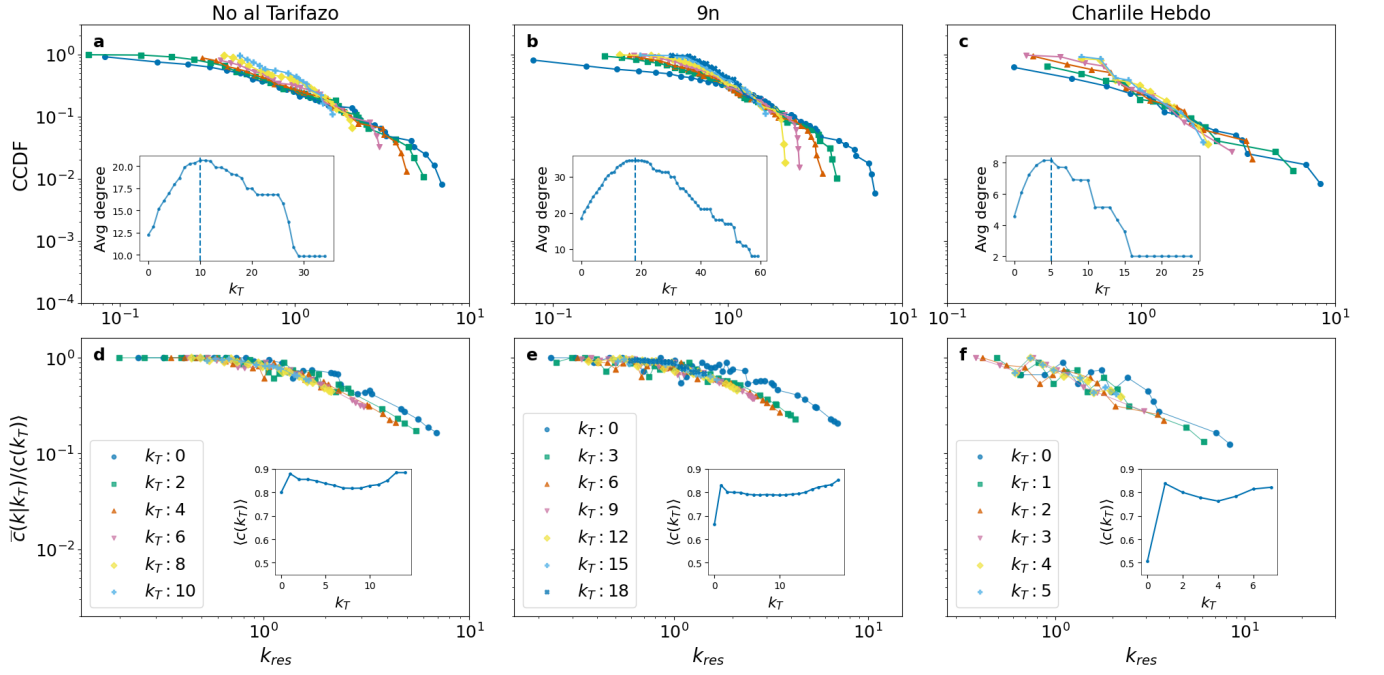


FIG. 3: **Complementary cumulative degree distribution and average clustering coefficient in the CTW.** The top row presents the complementary cumulative degree distribution function (CCDF) as a function of the rescaled degree, k_{res} , for various values of the degree threshold k_t . The inset illustrates how the average degree of the resulting networks increases with k_t . A blue dashed line indicates the cut-off point beyond which the results are excluded from further analysis due to finite-size effects forcing the average degree to decrease. The bottom row displays the spectrum of the average clustering coefficient rescaled by the average clustering coefficient of the corresponding subgraph as a function of the rescaled degree.

In Fig. 3a-c, we present the complementary cumulative degree distribution function (CCDF), which quantifies the probability that a node has degree no smaller than a certain value, for a selection of DTR subgraphs of each dataset at its CTW. For each curve, the x-axis shows the degree rescaled by the average degree of the corresponding subgraph, which we call k_{res} . The insets show how the average degree of the resulting subgraphs increases with the applied threshold k_t until it reaches a maximum (marked by a vertical dashed line), after which it decreases due to finite-size effects. We therefore stop the sequence of subgraphs at this maximum. Although the curves span a limited range of values, their collapse is clearly evident.

For each subgraph, we also calculated the clustering spectrum $\bar{c}(k)$ by averaging the local clustering coefficients $c_i = 2T_i/(k(k-1))$ of nodes in the same degree class k , where T_i is the number of triangles in which a node i participates. In Fig. 3d-f, we show the clustering spectrum of the subgraphs as a function of the degree rescaled by the average degree of the corresponding subgraph. Remarkably, we observe a clear collapse of the clustering coefficient onto a single functional form, across renormalisation steps for all three social movements. This collapse demonstrates the presence of a robust self-similar behaviour in the network structure over

multiple scales, where local patterns of organisation are replicated at progressively larger scales.

The slight variations in the range of k_t values reflect differences in network size and density among the three events, yet the fundamental self-similar behaviour remains consistent. Note that the original curve deviates slightly, indicating distinct statistical behaviour for the lowest-degree nodes.

The observed approximate scale-invariant behaviour suggests that the underlying organisational principles governing hashtag co-occurrence during critical mobilisation periods are largely independent of degree scale and persist across different levels of degree thresholding. Crucially, this property is specific to the CTW: as shown in Fig. SI.3, the CCDF and clustering spectrum curves fail to collapse for non-critical temporal windows across all three datasets, with ϵ^2 values substantially higher than those observed at the CTW, strongly suggesting that self-similarity is a distinctive structural signature of peak mobilisation.

D. Network embedding in hyperbolic space

The co-occurrence of clustering and the multiscale k -cut self-similarity in the network reconstructions of each

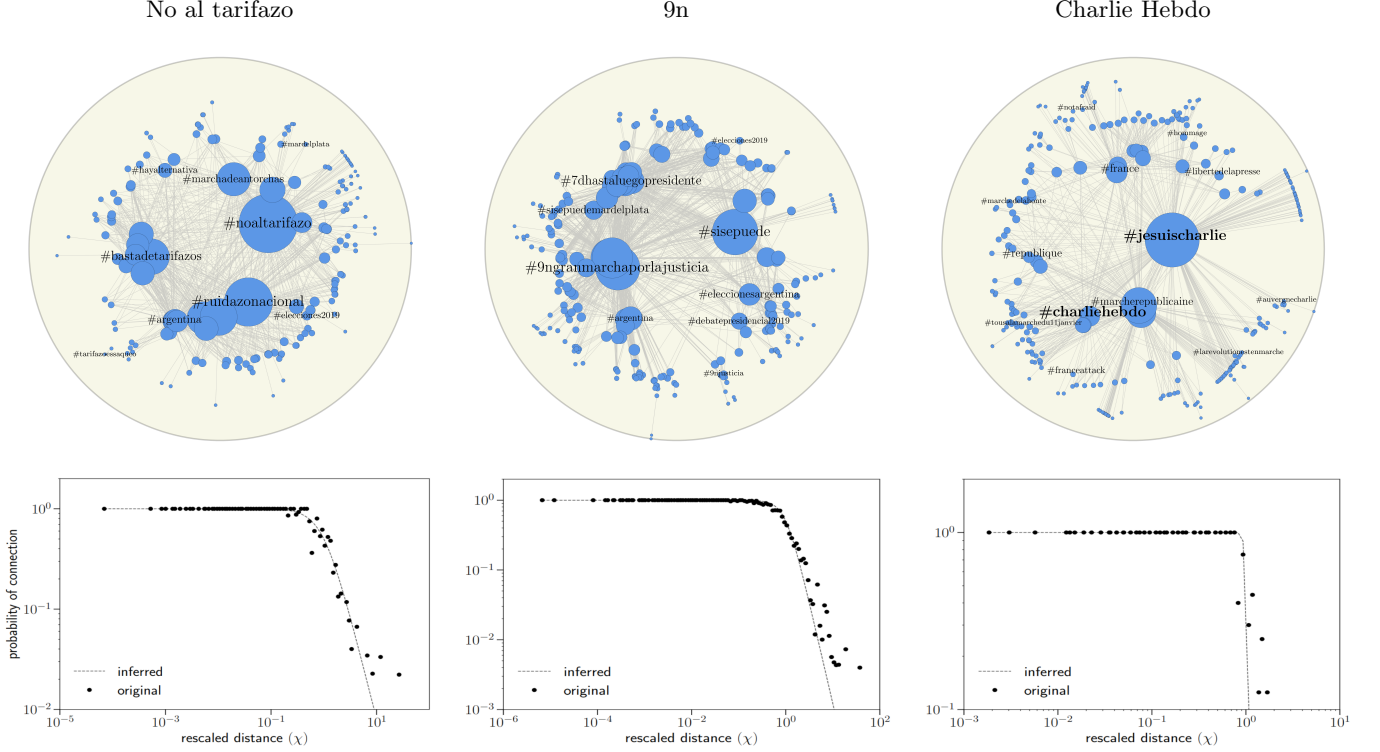


FIG. 4: Hyperbolic embeddings and connection probability profiles. The top row presents the two-dimensional projections of the hyperbolic embeddings obtained using the D-Mercator method for the filtered Critical Temporal Window networks of the three social movements. Node size is proportional to degree, and spatial proximity reflects the likelihood of connection in the underlying hyperbolic geometry. The bottom row displays the connection probability as a function of rescaled hyperbolic distance for each dataset. Black points represent the inferred model, whilst grey points correspond to the original network data. The excellent agreement between the model and empirical data demonstrates that the co-occurrence networks possess latent metric structures consistent with hyperbolic geometry, where connection probability decays smoothly with hyperbolic distance. From left to right: No al Tarifazo, 9n, and Charlie Hebdo datasets. Prominent hashtags such as **#noaltarifazo**, **#9ngrammarchaporlajusticia**, and **#jesuischarlie** are visible near the centre of each embedding, consistent with their high degree and central role in the mobilisation.

dataset at the CTWs points to connectivity linked to geometric constraints as a potential explanation for the observed symmetry [26]. Here we use the geometric $\mathcal{S}^1$ network model, which generates networks with clustering and multiscale k -cut self-similarity while nodes have a popularity coordinate directly controlling observed degrees and a similarity coordinate implemented as an angular position in a one-dimensional sphere or circle. Connections are formed with likelihoods that increase with popularity and decrease with angular distance. Equivalently, the model has a pure geometric version in the hyperbolic plane, the $\mathcal{H}^2$ model, where the probability of connection solely depends on hyperbolic distances [15].

We employed the model-based D-Mercator tool [19] to embed the CTW networks into the hyperbolic plane. Taking the network topology as input, D-Mercator calculates the popularity and similarity coordinates of the nodes that give the maximum congruence between the $\mathcal{S}^1/\mathcal{H}^2$ model and the observed topology.

Figure 4 displays the hyperbolic embeddings of the CTW networks for the three social movements alongside the comparison of their empirical and inferred corresponding connection probability curves. The upper row shows the two-dimensional hyperbolic embeddings, where node size represents degree and spatial proximity in the hyperbolic plane reflects the likelihood of connection. In each embedding, the seed hashtags—such as **#noaltarifazo** and **#ruidazonacional**, **#9ngrammarchaporlajusticia**, and **#jesuischarlie** and **#charliehebdos**—occupy central positions with the lowest radial coordinates, reflecting their high popularity and role as focal points of collective attention. Surrounding them, thematically related hashtags (e.g. **#sepuede**, **#marcherepublicaine**, **#libertéégalitéfraternité**) cluster at small angular distances, indicating high similarity, whilst more peripheral or loosely related hashtags are pushed towards the outer regions of the hyperbolic disc. The lower row presents the connection probability

as a function of rescaled hyperbolic distance, comparing the inferred model (dashed line) against the empirical network data (black points). For all three datasets, we observe excellent agreement between the inferred hyperbolic geometry and the original network structure. The connection probability decays smoothly with hyperbolic distance, closely following the expected behavior for networks embedded in underlying metric spaces. This successful embedding provides strong evidence that our co-occurrence networks can be explained by a latent geometric structure satisfying the triangle inequality. Moreover, the successful embedding across all three geographically and sociopolitically diverse movements further substantiates the universality of self-similar organisation in social mobilisation. Regardless of the specific context—whether protests against government policies in Argentina or solidarity marches following a terrorist attack in France—the underlying network structures exhibit consistent geometric properties that facilitate efficient information dissemination and rapid coordination among participants.

IV. DISCUSSION

In this work, we have investigated the emergence of self-similar structural patterns in social movements through the analysis of three distinct mobilisation events: the 9n justice reform protests and No al Tarifazo demonstrations in Argentina, and the Charlie Hebdo marches in France. By applying the DTR degree-thresholding renormalisation procedure to co-occurrence networks constructed from X (formerly Twitter) data, we have demonstrated that these highly correlated social phenomena exhibit clear signatures of self-similarity at critical periods. The observation that self-similarity emerges across all three datasets, despite their differing geographical and socio-political contexts, points towards a universal characteristic of social movements during critical periods.

Our methodology successfully identified Critical Temporal Windows (CTWs) for each movement. The convergence of multiple indicators—including the maxima in user and hashtag participation, the modular-to-nested transition (in both bipartite and co-occurrence networks), and the ϵ^2 test for features collapse—provides robust evidence that these CTWs represent distinct phases in the dynamics of social mobilisation. Notably, the identified CTWs align with the moments when digital activity translated into massive physical demonstrations, suggesting that the underlying network structures reflect fundamental organisational principles of collective attention.

The highly clustered nature of social interactions, driven by triadic closure mechanisms, is reflected in high levels of clustering. The clustering coefficient curves, as well as the degree distributions, collapse onto single curves obtained for the subgraphs generated by the DTR renormalisation procedure. Our interpretation is that this scale-invariant behaviour emerges from the co-

ordinated efforts of participants to achieve common goals within compressed timeframes.

Furthermore, our findings suggest that these networks possess a latent hyperbolic metric structure, which is thus applicable to social movement networks at critical periods. The successful embeddings using the D-Mercator method reinforces the notion that in these systems the likelihood of interaction decreases with an effective social distance, which combines similarity and popularity dimensions.

These results have important implications for understanding the organisational dynamics of social movements. The presence of self-similarity indicates that local patterns of coordination and information diffusion are replicated at progressively larger scales, facilitating the rapid amplification and synchronisation necessary for successful mass mobilisation. This hierarchical self-organisation may represent an optimal structural configuration for balancing information dissemination with maintaining network coherence during critical periods. Our results suggest that analysing other social phenomena characterised by peak correlation in attention and collective actions would be valuable for advancing our understanding of the fundamental principles governing criticality in social systems.

METHODS

A. Modular-to-nested transitions

For computing the modularity coefficient, we used the built-in modularity function in *networkx* [?], one of the most widespread Python libraries for graph analysis. It calculates the modularity of a certain graph $G = (V, E)$, being V the list of nodes -vertexes- of the graph and E the list of its edges, as defined in [29]:

$$Q = \frac{1}{2m} \sum_{ij} \left(A_{ij} - \frac{k_i k_j}{2m} \right) \delta_{g_i g_j},$$

where $\delta_{g_i g_j}$ is the Kronecker delta, which takes value of 1 if $g_i = g_j$ and 0 otherwise, $m = |E|$ is the number of edges of the graph, A is the adjacency matrix of G : A_{ij} is equal to the weight if there is an edge connecting node i to node j , and $A_{ij} = 0$ otherwise.

To quantify nestedness, we use the NODF (Nestedness based on Overlap and Decreasing Fill) [30, 31]. By doing so, we quantify the extent to which the interactions of a node form subsets of the interactions of the higher-degree ones. Specifically, in undirected networks NODF is calculated as follows:

$$\text{NODF}_{uni} = \frac{2}{n(n-1)} \sum_{\substack{i < j \\ k_i \neq k_j}}^n \frac{O_{ij}}{\min(k_i, k_j)} \quad (1)$$

The quantity $O_{i,j} = \sum_{k=1}^n A_{ik}A_{jk}$ measures the number of common neighbours between each pair of nodes i and j using the overlap matrix $O_{i,j}$.

In this way, we obtain the average, over all pairs of nodes, of the overlap normalised by the lesser degree of the pair. Then, normalising the overlap of neighbours by the smaller degree captures hierarchy, because it tests whether the neighbours of the less-connected node form a subset of the more-connected ones. For example, if all of the smaller node's neighbours are also linked to the larger one, the ratio equals 1 (perfect nestedness). While if only some are shared, it lies between 0 and 1. Then, the average of this measure across all node pairs quantifies how much the network approaches the ideal nested (hierarchical) structure, where low-degree nodes are entirely embedded into the neighbourhoods of high-degree ones. Prior to computing nestedness, the weighted bipartite and co-occurrence networks were binarised by setting all non-zero edge weights to one. For both the unipartite co-occurrence and bipartite user-hashtag networks, Eq. (1) was applied to the full $n \times n$ binary adjacency matrix, where $n = R + C$ in the bipartite case, with R users and C hashtags.

B. ϵ^2 test to measure distribution's collapse

To quantify the collapse between the original distribution and those obtained after the procedure of degree-thresholding renormalisation, we compute the distance between them by means of the ϵ^2 test. In this way, we quantify the difference between the original distribution and the subsequent renormalised ones. For that endeavour, we leverage the algorithm presented in [26] to calculate the values of $\epsilon_{\hat{c}(k)}$:

$$\epsilon^2 = \frac{1}{n+1} \cdot \frac{1}{n_{bins}} \sum_{k_T} \sum_{i=1}^{n_{bins}} \left(\frac{\text{bin value}_i^{org} - \text{bin value}_i^{k_T}}{\text{bin value}_i^{org}} \right)^2,$$

where $n_{bins} = 20$. We performed the ϵ^2 test to find the time-window where the generated networks collapse - present self-similar behaviour- sweeping through all time windows and looking for the occurrence of a sudden drop.

On the other hand, as the division in TW's is a way to discretise the continuous time range of the timespan the dataset covers, it can cause anomalies in certain TWs. To avoid these noise errors, we decided to calculate the moving average of the last $N = 3$ TW's. Following this procedure was easy to find the right TW, identifying the TW with the lowest $\epsilon_{\hat{c}(k)}$.

C. Geometric model and network embedding

To investigate further whether the observed self-similarity at CTWs corresponds to an underlying geometric structure, we employed D-Mercator [19], a model-based embedding method that maps real networks into the $(D + 1)$ -dimensional hyperbolic space base on the $\mathbb{S}^D$ model [14]. D-Mercator implements a maximum-likelihood inference procedure that assigns each node two hidden coordinates: a popularity coordinate corresponding to a radial position in the latent hyperbolic space that correlates with node degree, and similarity coordinates corresponding to angular positions in a D -dimensional sphere that determine community structure. For simplicity, we used the two-dimensional version, where every nodes is characterized by a popularity variable or hidden degree κ_i and an angular position θ_i in the one-dimensional sphere representing the similarity space. In $D = 1$, D-Mercator infers the popularity-similarity coordinates of the $\mathbb{S}^1$ model [14] or, equivalently, the $\mathbb{H}^2$ formulation [15] in the hyperbolic plane, and has been shown to accurately reproduce complex network topologies whilst remaining computationally tractable. The method assumes that the connection probability between nodes i and j follows:

$$p_{ij} = \frac{1}{1 + \left(\frac{R\Delta\theta_{ij}}{\mu\kappa_i\kappa_j} \right)^\beta}, \quad (2)$$

where R is the radius of the similarity circle, $\Delta\theta_{ij}$ is the angular distance between the nodes, μ controls the average degree, and β controls the level of clustering quantifying the coupling between the network topology and the underlying geometry.

Prior to embedding, we filtered each CTW network to remove hashtags tweeted by fewer than N users, as low-activity nodes introduce noise that degrades inference quality. The threshold N was selected based on network size and density: after testing $N \in \{3, 5, 7, 10\}$, we determined that $N = 3$ for *No al tarifazo* and *Charlie Hebdo* and $N = 5$ for *9n* provides the optimal trade-off between noise reduction and preservation of structural features (see Table SI.1) for filtering statistics).

The quality of each embedding was assessed by comparing the empirical connection probability $P(\chi)$ as a function of rescaled hyperbolic distance with the theoretical prediction from the inferred model. High-quality embeddings exhibit smooth decay of $P(\chi)$ with distance, indicating that the network structure is well-explained by the latent hyperbolic geometry.

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